# Saurabh Verma

**Associate @ Nomura | 3+ Years Experience | IIT Bombay'22 | FRM I Qualified (Aug'25)**

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## EDUCATION

| Degree | Institute | Department | Year |
|---|---|---|---|
| Graduation (B.Tech.) | Indian Institute of Technology, Bombay | Mechanical Engineering | 2018-2022 |

## PROFESSIONAL SUMMARY

Technical professional with 3+ years of experience in data-driven problem solving, large-scale system development, quantitative and statistical analysis, and model validation. Proven track record in designing scalable solutions, optimizing complex processes, and delivering high-impact tools in dynamic financial environments. Strong foundation in mathematical modelling, technology integration, and innovative solution design, with a focus on accuracy, efficiency, and continuous improvement.

## SKILLS & CERTIFICATIONS

**Development :** Python, C++, Java, SQL, Low & High Level Design, OOPS, React.js, Java, Git, CI/CD, Flask

**Data Science :** Pandas, NumPy, ML/DL, LLM, Seaborn, MatplotLib, PyTorch, sci-kit-learn, Tensorflow, Keras

**Finance :** Quantitative Analysis, Financial Markets and Products, Valuation and Risk Models, Risk Management

**Certifications :** FRM Level I, Machine Learning A-Z, Python (Specialization of 5 Courses), SQL, JavaScript [Coursera/IBM]

## PROFESSIONAL EXPERIENCE

**Nomura | Associate | Model Validation Group (Quants)** [Feb'25 – Present]

**Market Risk Model Validation Tasks**
- Performed validation of **FRTB** models, ensuring alignment with **Basel III** standards and maintaining model robustness and accuracy
- Built an **LLM-based** compliance tool using **LangChain, Anthropic-3.5**, and **RAG** technique for validation reports' regulatory checks
- Used **asyncio** to achieve **10x faster** processing and generated ratings from comprehensive report analysis based on SS1/23 principles
- Gained deep understanding of **VAR** models, including modeling assumptions, risk factor decomposition, and model limitations

**Bank of America | Software Developer | Equity Derivatives' Lifecycle Management** [Jun'22 – Feb'25]

**Trade Life-Cycle Evaluation Service**
- Led migration from Java to Python for a legacy application, applying **high-level architectural** knowledge and **low-level design** skills
- Established communication between Relational/ Object Oriented databases & backend services using **AMPS messaging** channels
- Automated & optimized the daily evaluation of 50,000+ Instruments for 15,000+ Barrier breaches, ensuring scalability and efficiency
- Implemented **10+ services** for data loading, load balancing, **life-cycle evaluation**, and booking **facilitating** all **trading markets**
- Automated daily booking of 250+ coupon/ termination payments, and barrier breaches daily, enhancing overall process efficiency

**Barrier Options Monitoring Dashboard**
- Independently developed **end-to-end** application, integrating database, via **SQL**, **Flask** services, and a responsive **React** front-end
- Established **REST APIs** with Flask, incorporating Kerberos authentication for seamless & secure data communication across services
- Orchestrated a robust **CI/CD** pipeline using Jenkins, Bitbucket, and Ansible to enable streamlined and automated deployment

## KEY ACADEMIC PROJECTS

**Sentiment Analysis**
- Analyzed 515k+ hotel reviews data using word clouds, histograms, bar charts, heat maps and other visualization methods for insight
- Achieved an accuracy of 89% on testing data using **Random Forest Classifier** with Bag of Words technique for sentiment analysis

**Recommendation System**
- Developed a content-based recommendation system from raw data of 45,000+ movies across 85+ different languages used globally
- Utilized Regular Expression for data cleaning and implemented **cosine similarity** on tf–idf vectorized data of the combined features

**Credit Card Fraud Detection**
- Developed a fraud detection system using **Random Forest** and **GaussianNB**, achieving an F1 score of 0.85 on fraudulent transactions
- Addressed severe class imbalance with SMOTE and Random Over Sampler, improving fraud recall to 0.99 for minority class detection

**Object Detection**
- Compiled extensive construction-related video datasets and annotated them using **Computer Vision** Annotation Tool (CVAT)
- Built a robust pipeline for CVAT data preprocessing, converting annotations into **PyTorch** tensors to support effective model training
- Implemented **transfer learning** on a **Faster R-CNN** model with PyTorch, for precise and accurate detection of construction objects

## ACHIEVEMENTS & ACCOMPLISHMENTS

- Cleared **FRM** (Financial Risk Manager) Level I, showcasing expertise in finance and quantitative risk modeling (2025)
- Recognized with Platinum and Bronze awards for exceptional performance at Bank of America (2024)
- Among top 4 teams within APAC region in Machine Learning Code Challenge conducted by Bank of America globally (2023)
- Recipient of the Kishore Vaigyanik Protsahan Yojana (**KVPY**) fellowship by the Government of India (2018)
- Qualified for Indian National Mathematics Olympiad (**INMO**) along with top 835 students across India (2016)